NVIDIA Jetson Orin Nano Super

A branch that argues against its own hardware

Compute	6× A78AE + 1024-core Ampere
Rated	67 TOPS at 25 W
GPU utilisation	**none, by 13 500×**
Electricity	EUR 44/y against the Pi 5's EUR 13
What survives	data sovereignty, multimodal fusion
Cost	∼EUR 320

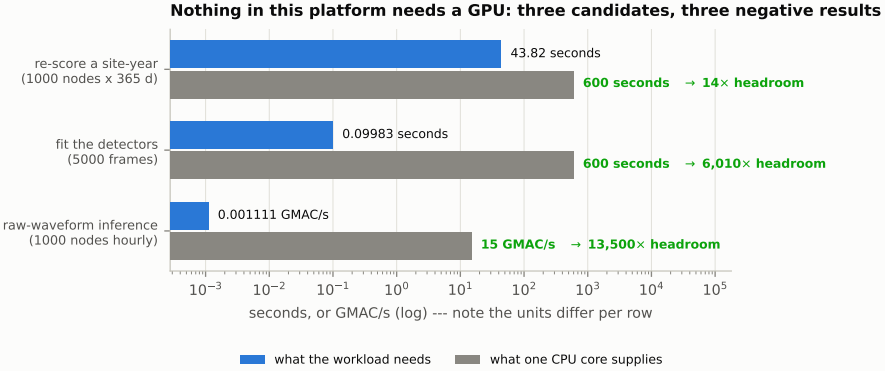
Two earlier branches already found that accelerators do nothing for feature-vector scoring — Alif on an Ethos-U55, the Pi 5 on a Hailo-8L. A third negative result would be a footnote.

So the brief was inverted: is there a workload here where a GPU is the difference between *possible and impossible*, rather than between fast and faster?

Never run on a Jetson. And `torch` is absent, so the one GPU-relevant number is missing — asserted by a test, not forgotten.

Branch `hw/jetson-orin-nano`. 13 tests. Figures from `results/workload_study.json`.

Three candidates, three negative results



Re-scoring a 1000-node site's *entire year* — 8.76 million frames — takes 44 seconds of CPU. The classical detectors fit in a tenth of a second. Raw-waveform inference at hourly cadence is four orders of magnitude below one core.

The study records where candidate 3 would flip: **3 750 inferences/s**, i.e. 13.5 million nodes at hourly cadence.

Jetson Orin

The honest weak spot, and the study says so.

Candidate 2's fits are closed-form and SVD, not gradient descent. A Deep SVDD or a 1D CNN is the one training workload a GPU would change, `torch` is not installed here, and so **that number is absent rather than guessed**.

A test asserts `torch_available is False` and fails if torch ever appears, so the gap cannot be quietly forgotten.

What survived: neither justification is about speed

1. Data sovereignty — implemented.

A plant that will not send vibration data off site cannot have its models trained in a cloud. Pharmaceutical, defence, and increasingly any site whose legal team has read a data-processing agreement.

That makes the *site* the place where training happens, and a site that trains needs more than a gateway. It is the only justification here about *where* computation happens rather than how fast — which is why it survives an argument about throughput.

`train_onsite.py` trains from the gateway's own queue, read-only, because those frames still have to go upstream.

2. Multimodal fusion — not written.

Vibration + acoustic + motor current + thermal imagery on one asset does not fit a 63-float feature vector at all, and the imagery alone rules out the MCU and gateway tiers. It is the real GPU case.

Three accelerators in this lineup — Ethos-U55, Hailo-8L, Ampere — and all three are justified by models that do not exist yet.

That is the honest state of the art here. A stub for any of them would make the platform look otherwise.

So what: Buy it when a legal department makes on-site training mandatory. That condition will arrive before the multimodal model does.

Two bugs, both “checks that cannot fail”

The two-window version, which was wrong:



calibrated AND verified on the same window: the rate equals the target by construction

The three-window version, split by TIME not at random:



time → a later window the threshold has never seen

and the gate is a 3-sigma POISSON band on the alarm COUNT, not a multiple of the rate: on 175 frames at a 1 % target, 1.8 alarms are expected and noise alone spans 0-6.

The circular check. Fit on 75 %, calibrate the threshold on the last 25 %, then measure the false-alarm rate on *that same* 25 %.

That equals the target by construction. The check could never fail, and a deliberate 4× drift sailed through it reporting exactly 0.010 — which is how the bug announced itself: **a suspiciously exact number**.

The statistically wrong gate. The replacement refused if the verification *rate* exceeded the target by 4×. On 175 frames at 1 %, the expected count is 1.8 and Poisson noise alone spans 0–6 at 3σ — so it refused stationary data a third of the time.

In a plant that means re-running training until it passes, which **selects for whichever window looked quiet**: a biased model, reached by a procedure that feels like diligence.

The refusals are most of the value

Gate	Limit	The failure it prevents
Minimum frames	400	A 63-feature covariance estimate needs ∼189, and the 50/25/25 split gives the fit window half
Minimum span	**7 days**	**The one that bites.** A commissioning engineer with a day on site has plenty of frames and nowhere near enough span; a detector that has never seen a Monday start-up alarms on the first one
Unscoreable frames	excluded	A detector trained on clipped acquisitions learns the artefacts, then fails to alarm on the real thing because the real thing is *less* extreme
`SPEED_UNSTEADY`	excluded	Their high-order content is smeared, so training on them teaches the detector that smearing is normal
Feature-spec mismatch	excluded	Mixing spec versions in one fit mixes two different feature layouts

`1 of 3 accepted` on a synthetic site: one good channel, one refused for a 0.3-day span, one refused for a population that moved.

The span gate will be argued about on site, and it should be: seven days is a *minimum*, not a target. A monthly duty cycle needs a month.

Verdict

Workload study, 3 candidates	all negative
On-site training with gates	13 tests
Three-window time split	✓
Poisson verification gate	✓
Queue not consumed	✓
Run on a Jetson	no
`torch` installed	no
GPU used at all	no
Multimodal fusion	no model exists
Rollout / rollback	not written

Legacy Jetson Nano (2019): end-of-life, last JetPack is 4.6 on Ubuntu 18.04. Everything here runs on it in principle; deploying an unpatched internet-connected Linux to a plant network is a finding, not a trade-off.

The output of this branch is a recommendation against its own hardware.

Buy the Pi 5. Revisit when a legal department makes on-site training mandatory, or when a multimodal model exists that genuinely needs a GPU.

And keep the training pipeline either way — the gates, the time split and the Poisson band are worth having on whatever box the site ends up with.

The EUR 31/year of extra electricity is the cheapest part of being wrong, which is exactly why nobody examines it. The study exists so somebody does.